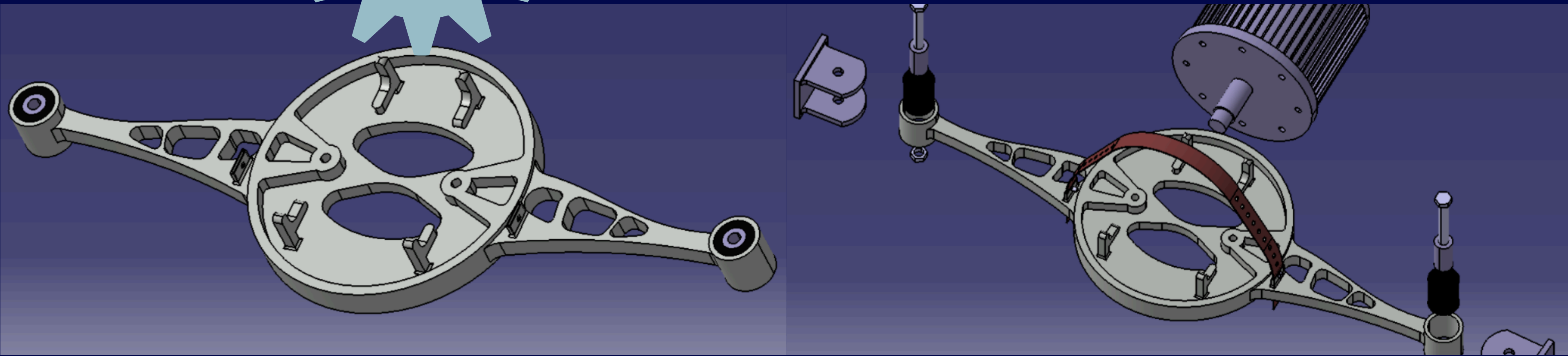


Additive Manufacturing



PORTFOLIO

SUDEEP GUDURU

Mechanical Engineer

Additive Manufacturing | DFAM | CAD | FEA

Design | Analysis | Manufacturing

sudeepchowdary27@gmail.com

+91 7816057828

CONTENT

01 ABOUT ME & SKILLS

09-15 PROJECT 2: PCB ENCLOSURE

02-08 PROJECT 1: EV MOTOR MOUNT

21 CONTACT

ABOUT ME

I am a Mechanical Engineering graduate with a strong interest in Additive Manufacturing, Design for Additive Manufacturing (DFAM), CAD and Engineering Analysis.

I enjoy solving mechanical design problems through lightweighting, design optimization and structural validation. My project work has given me experience in developing mechanical components, refining designs through multiple iterations, and evaluating their structural performance using FEA.

I am particularly interested in building a career in Additive Manufacturing, with a focus on DFAM, process development, lightweight structures and advanced manufacturing applications.

TECHNICAL SKILLS

Design & CAD

CATIA V5
SolidWorks
3D Modeling
Mechanical Design

Engineering Analysis

ANSYS
Static Structural FEA
Design Optimization
Lightweighting

CAREER FOCUS

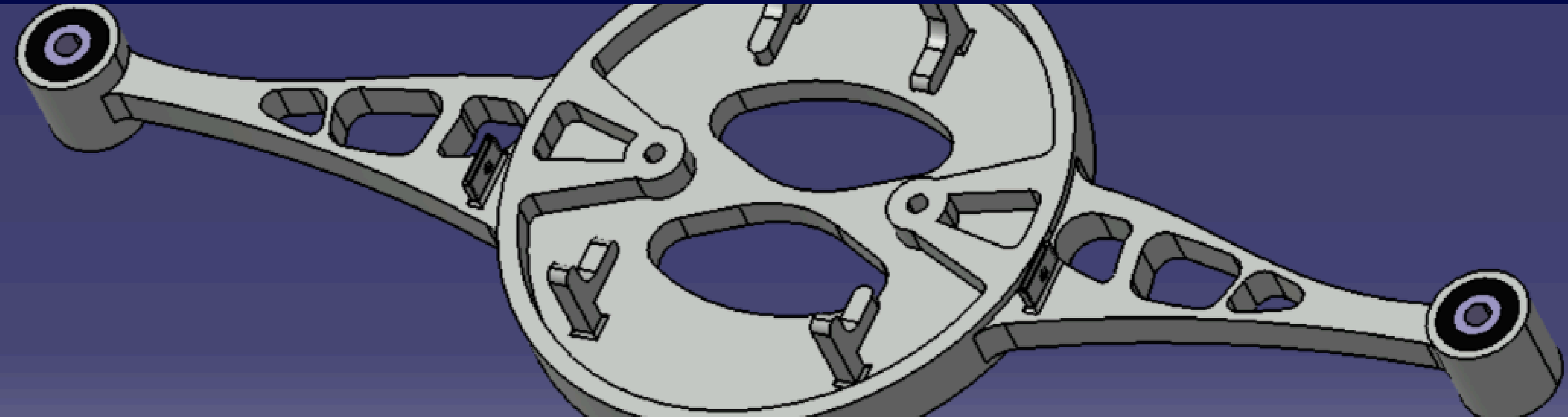
Additive Manufacturing • DFAM •
Lightweight Design • Process
Development



PROJECT 1

LIGHTWEIGHT EV MOTOR MOUNT

DESIGN | LIGHTWEIGHT | CAD | FEA | DFAM



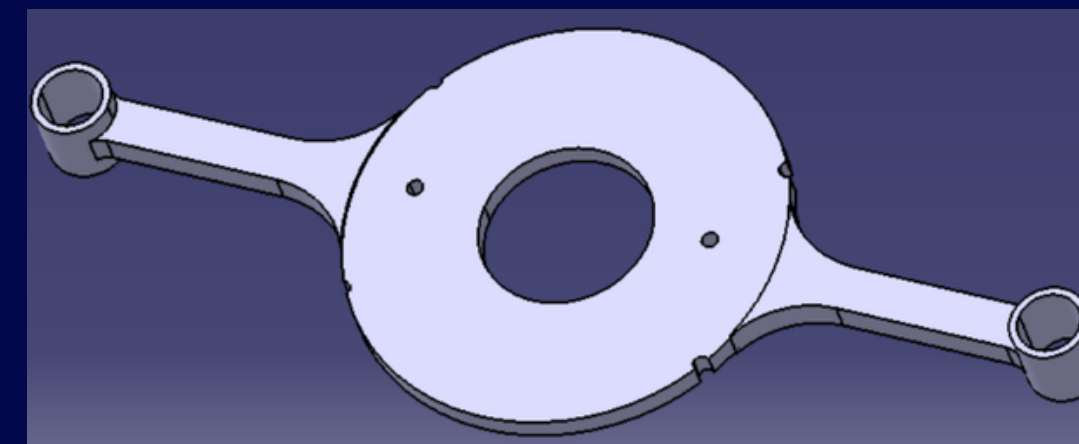
PROJECT OBJECTIVE

Design and Developed a lightweight EV motor mount focused on improving material efficiency while maintaining structural integrity and secure motor retention.

DESIGN AND DEVELOPMENT

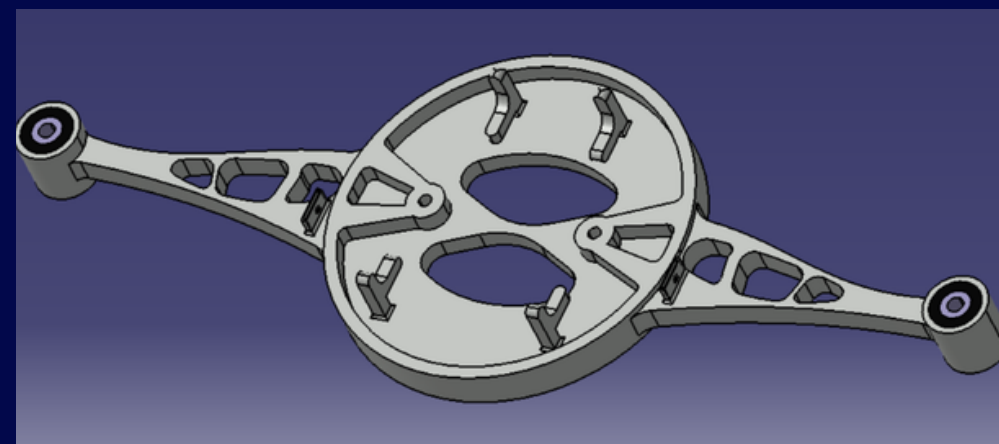
DESIGN EVOLUTION

From concept to final design



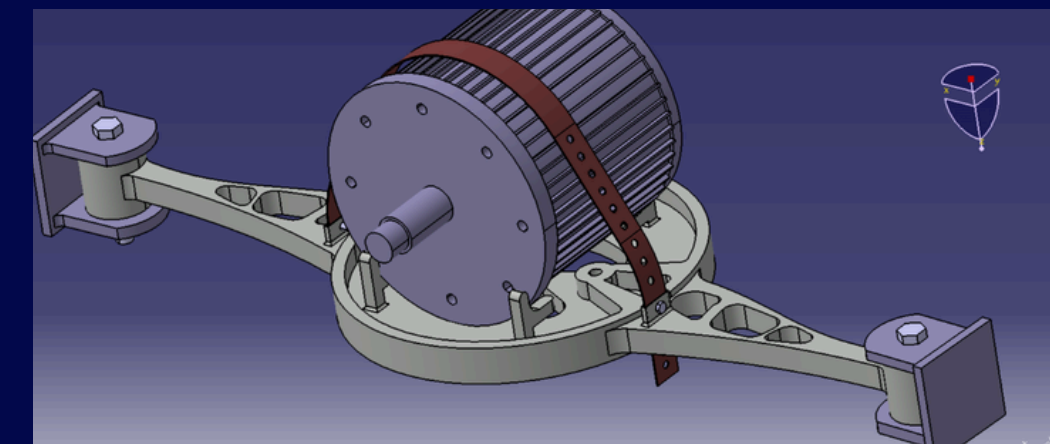
Initial Concept

Developed the basic motor mounting geometry and established the primary mounting interfaces.



Structural Development

Refined the geometry by introducing lightweight features while retaining critical structural regions.



Final Assembly

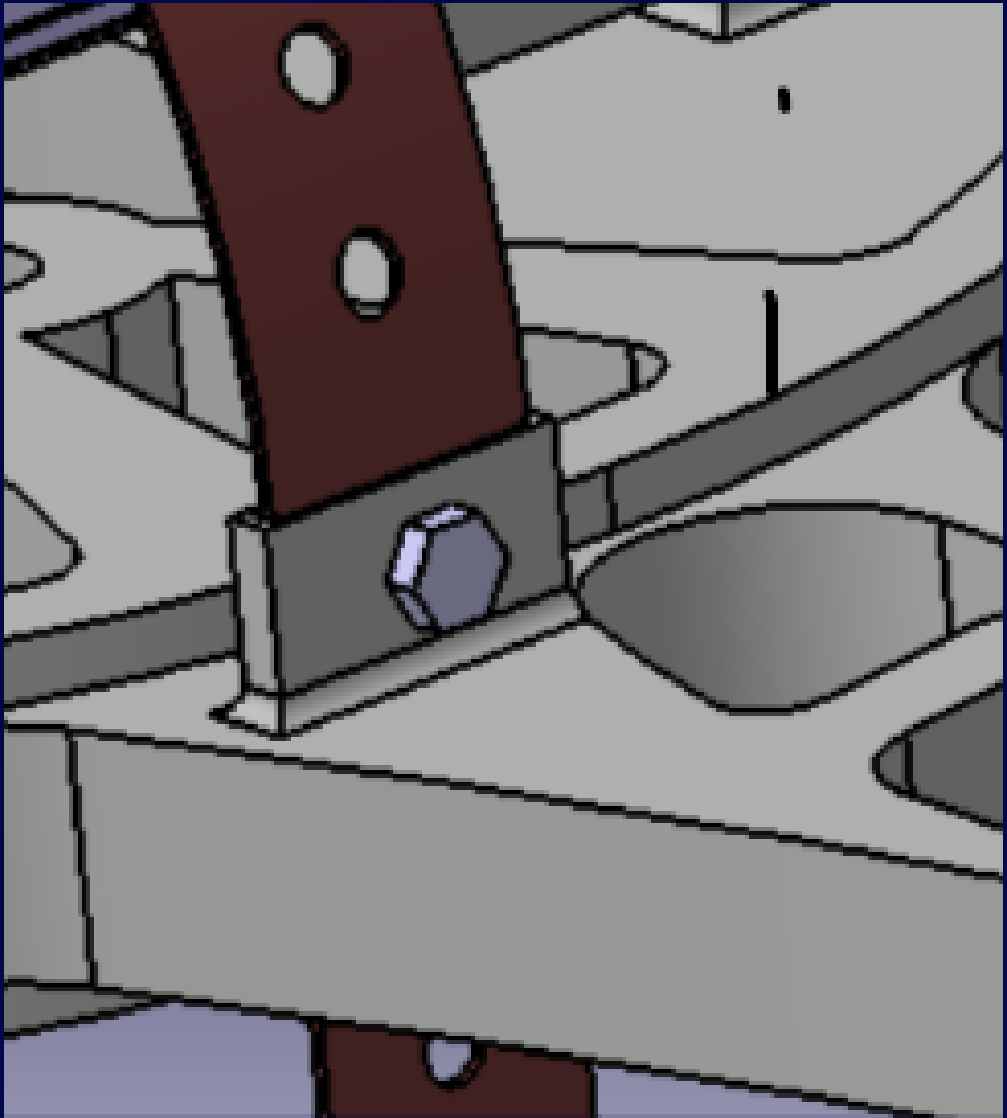
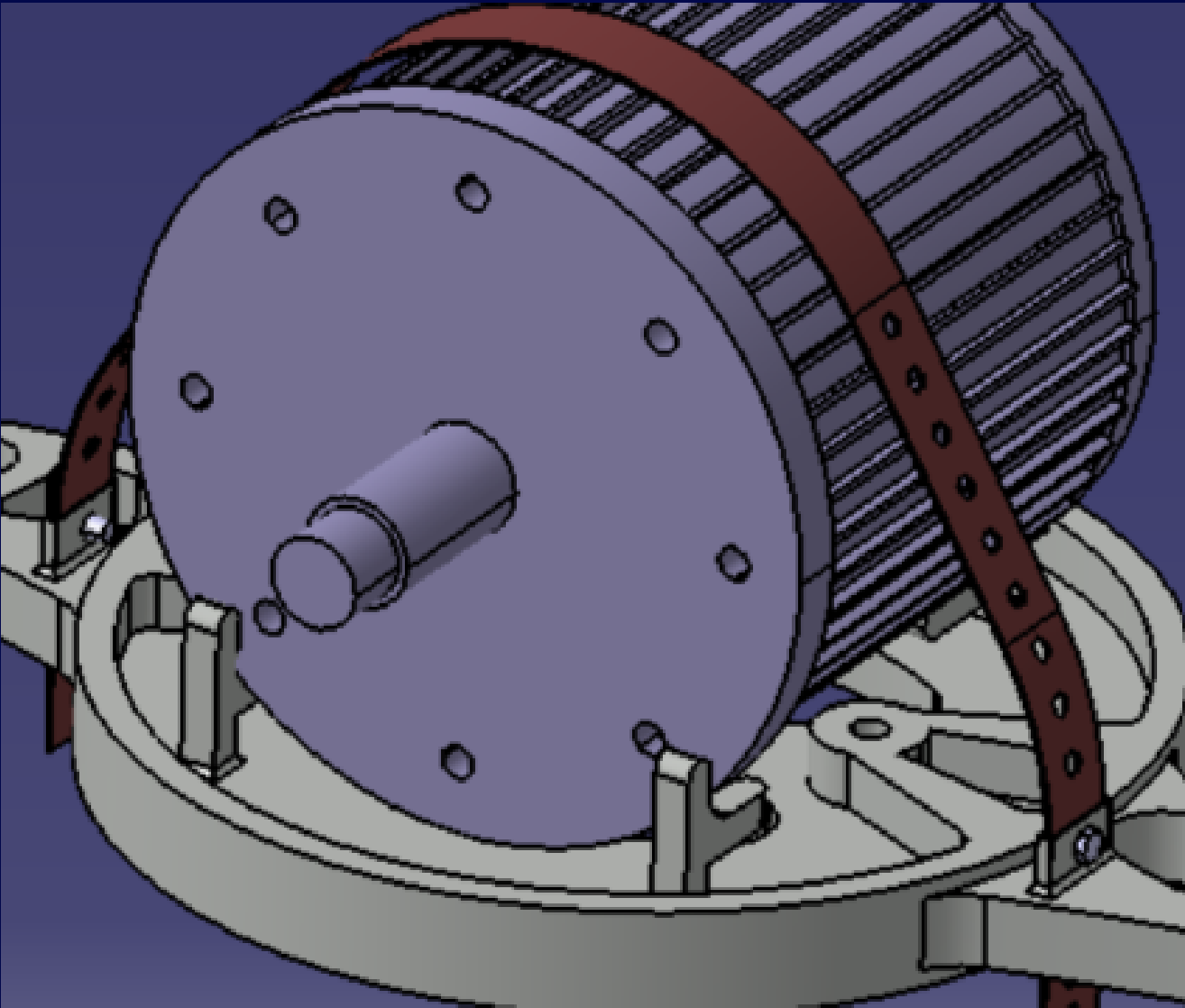
Integrated the motor with the redesigned mount and incorporated the retention system for secure positioning.

DESIGN APPROACH

1. Identified critical mounting and load bearing regions.
2. Refined the geometry to reduce unnecessary material.
3. Improved the structural layout while maintaining motor mounting interfaces.
4. Developed the final geometry with manufacturability in mind.

MOTOR RETENTION SYSTEM

Secure | Adjustable | Integrated



01 - SECURE RETENTION

Provides additional restraint to prevent unwanted motor movement during operation.

02 - ADJUSTABLE TENSION

The belt arrangement allows controlled tensioning around the motor.

03 - STRUCTURAL INTEGRATION

Retention features are integrated into the mounting structure without significantly altering the overall design.

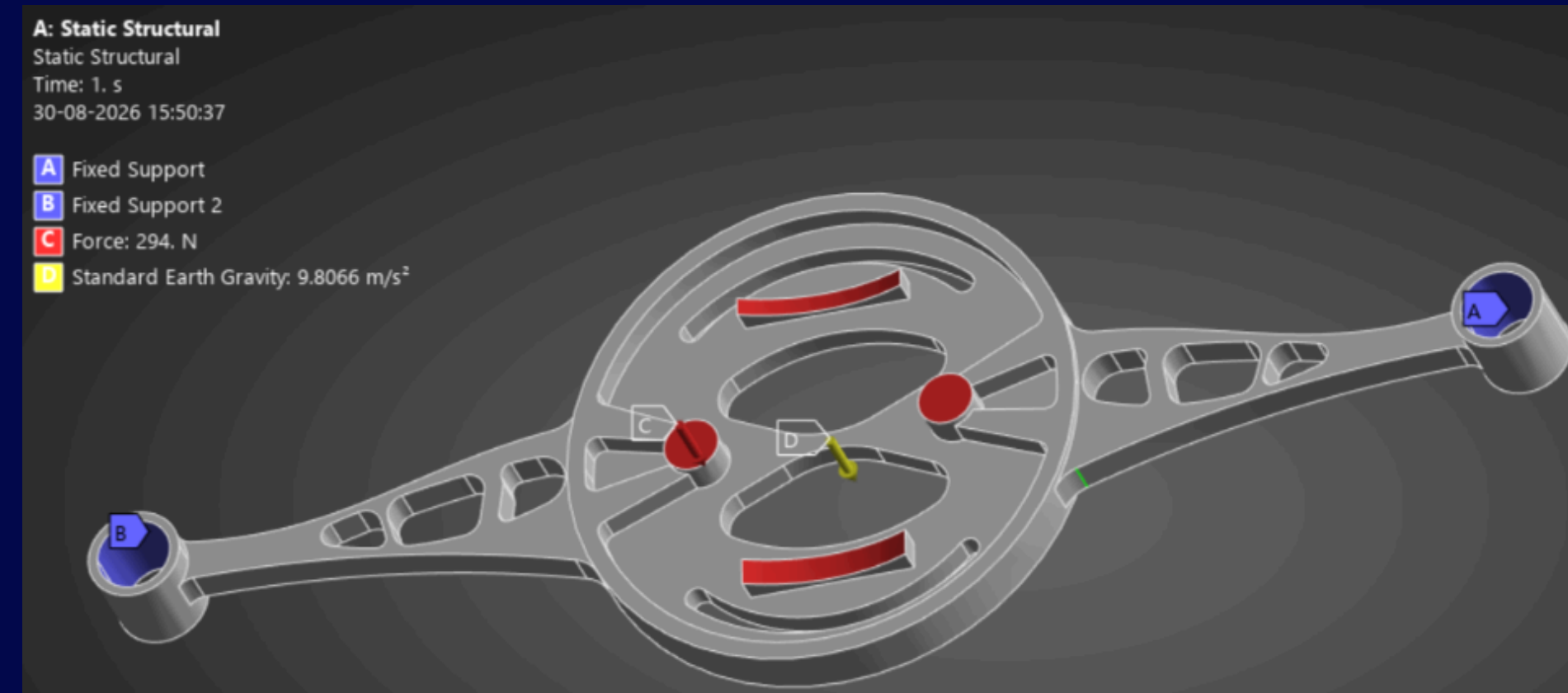
RETENTION APPROACH

A belt-based retention system was incorporated to securely retain the motor within the mounting structure. The arrangement provides additional restraint while maintaining accessibility and integration with the lightweight mount.

FEA & STRUCTURAL VALIDATION

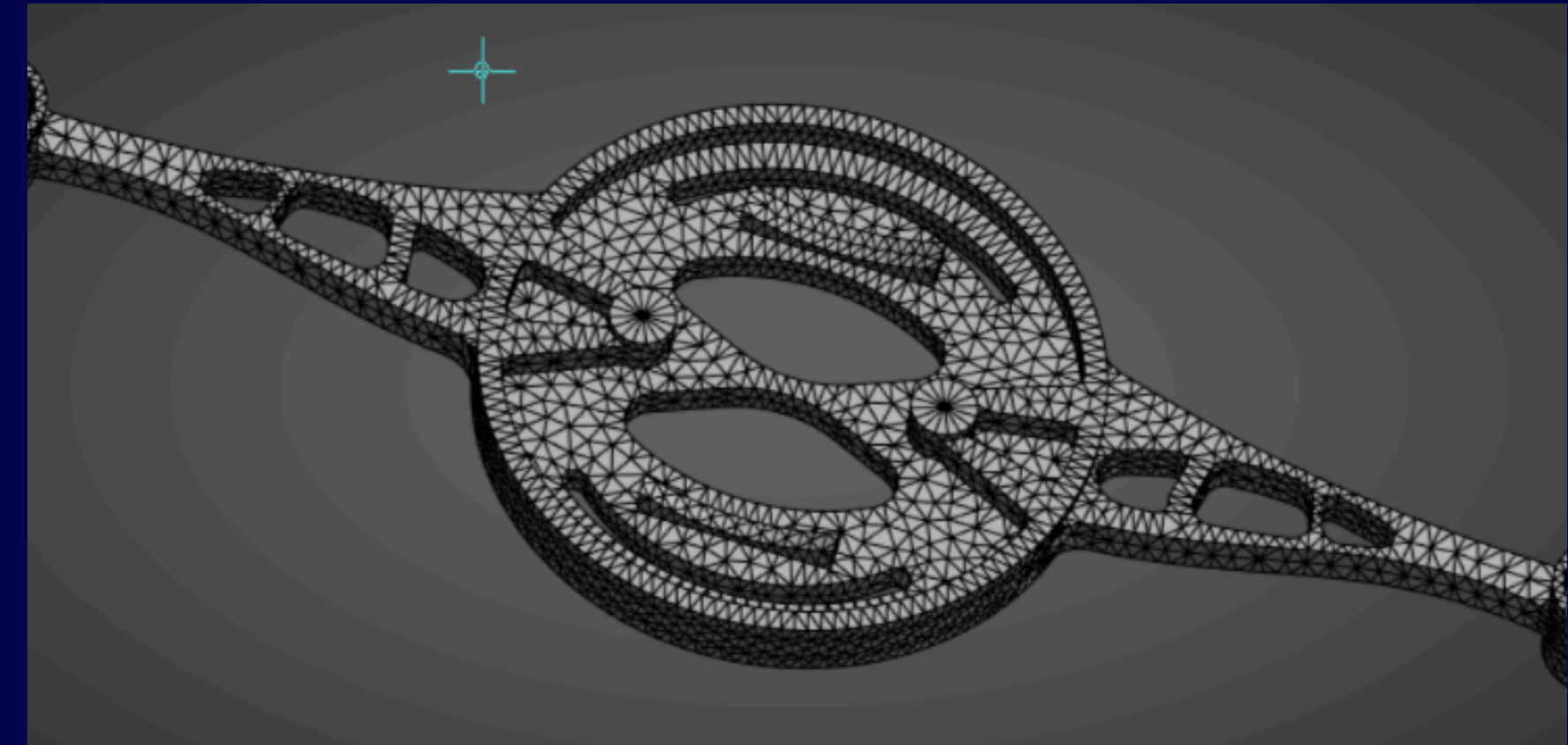
Validation of the lightweight motor mount under static loading

Boundary Conditions



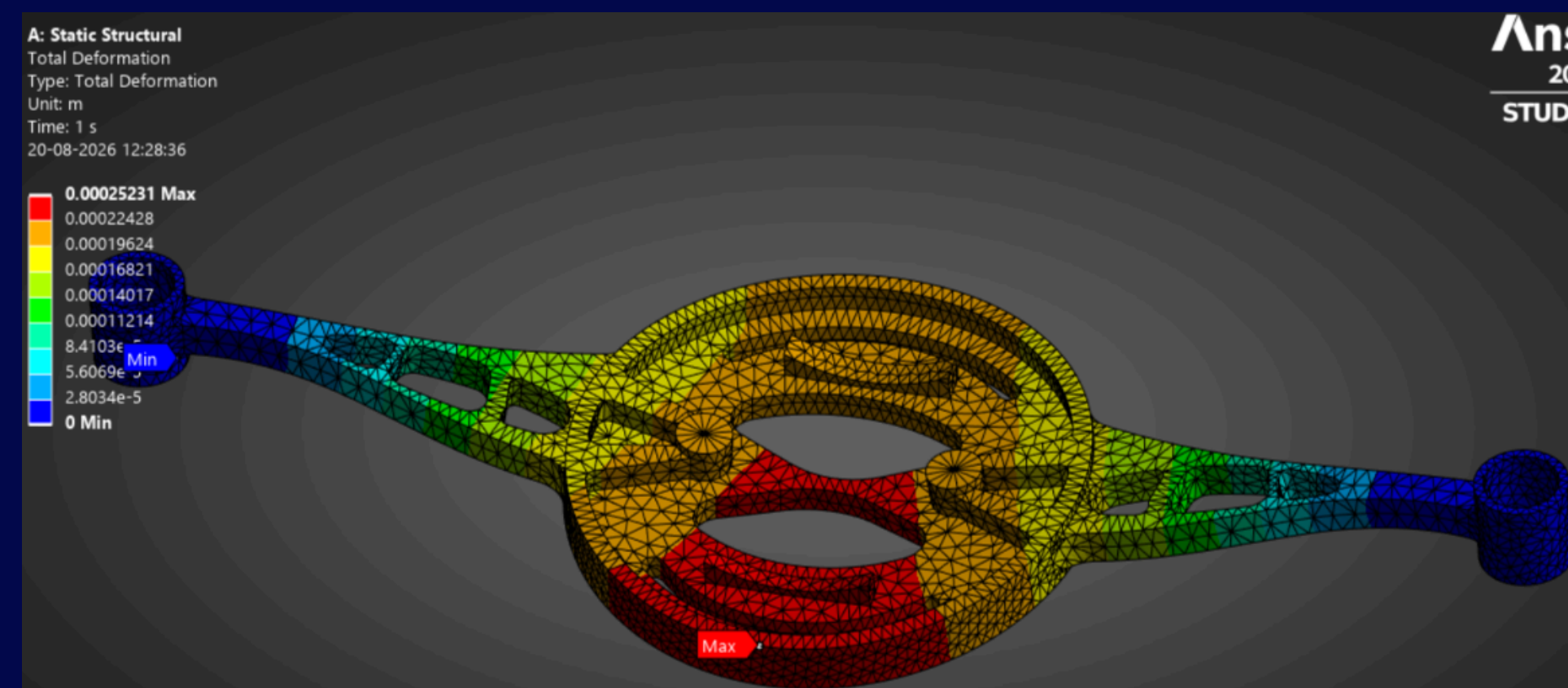
1. Fixed supports at the mounting points.
2. Applied force: 294 N
3. Standard Earth Gravity: 9.8066 m/s²

Mesh Generation

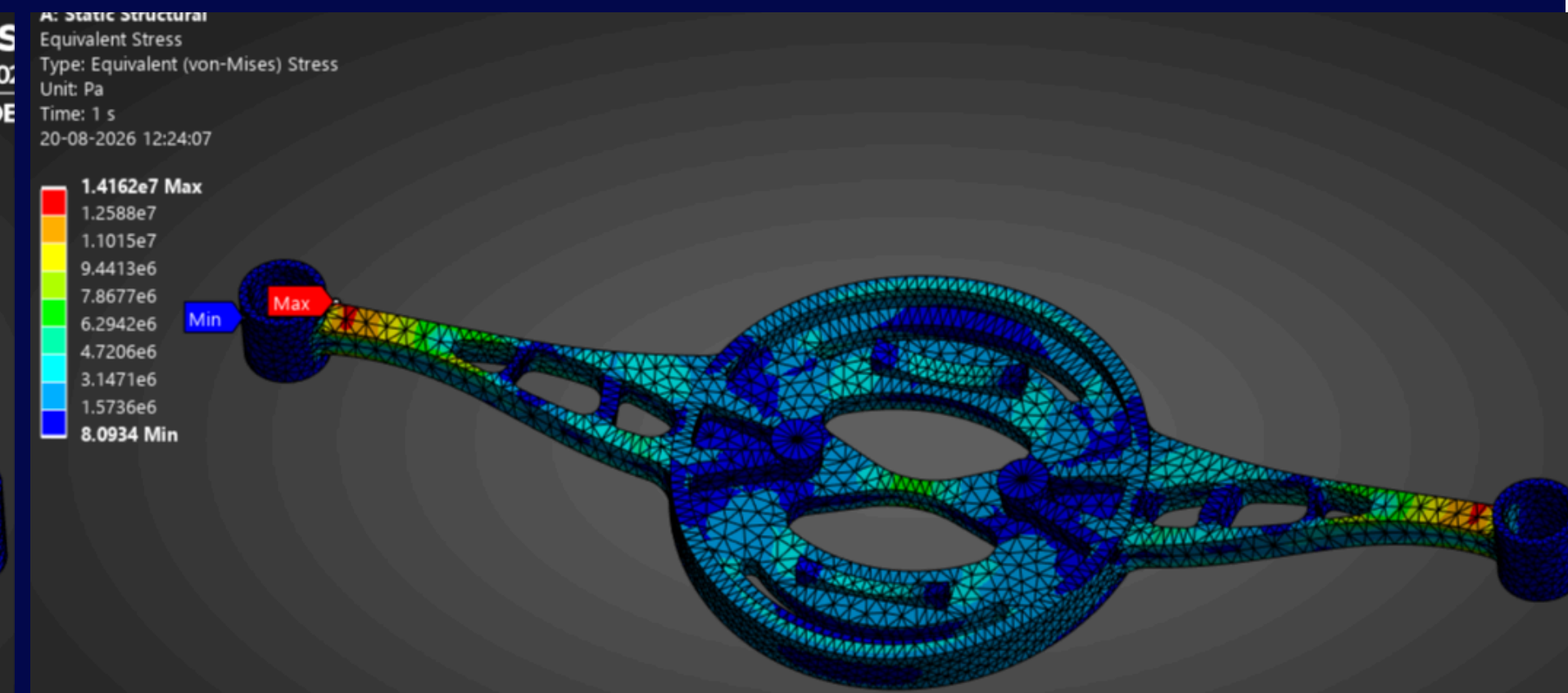


1. Finite element mesh generated over the complete mount geometry.
2. Mesh captures the lightweight structural features and load - bearing regions.

Total Deformation



Equivalent Stress



1. Maximum deformation: ≈ 0.252 mm
2. Deformation remains concentrated around the central loaded region.

1. Maximum von-mises stress: ≈ 14.16 MPa
2. Highest stress occurs near the mounting/load-transfer regions.

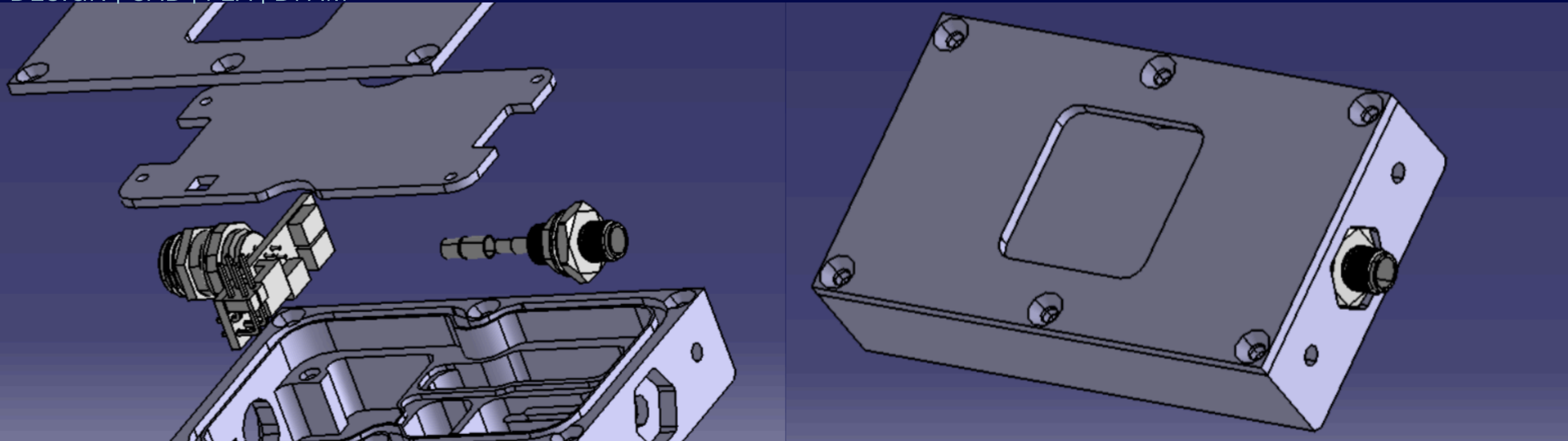
AM-ORIENTED DESIGN

1. Lightweight geometry with material-efficient features
2. Reduced unnecessary material through structural optimization
3. Geometry designed with Additive Manufacturing in mind
4. Integrated features reduce the need for multiple components
5. Complex lightweight geometry is well suited to metal AM processes

PROJECT 2

PCB ELECTRONICS ENCLOSURE DESIGN & THERMAL ANALYSIS

DESIGN | CAD | FEA | DFAM

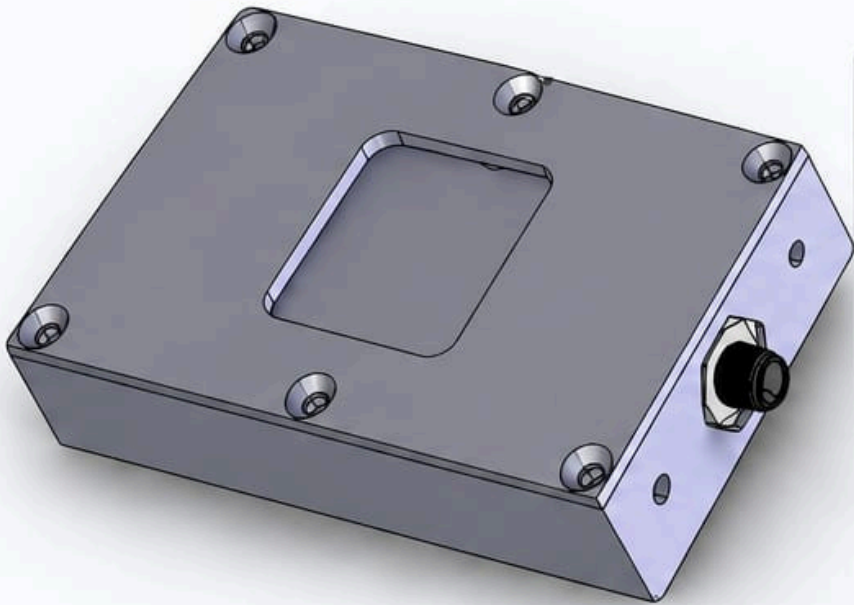


PROJECT OBJECTIVE

1. Designed a compact enclosure for a PCB and electronic components.
2. Integrated mounting features, PCB interfaces, IC/heatsink arrangement, and enclosure details.
3. Evaluated thermal behavior using ANSYS.

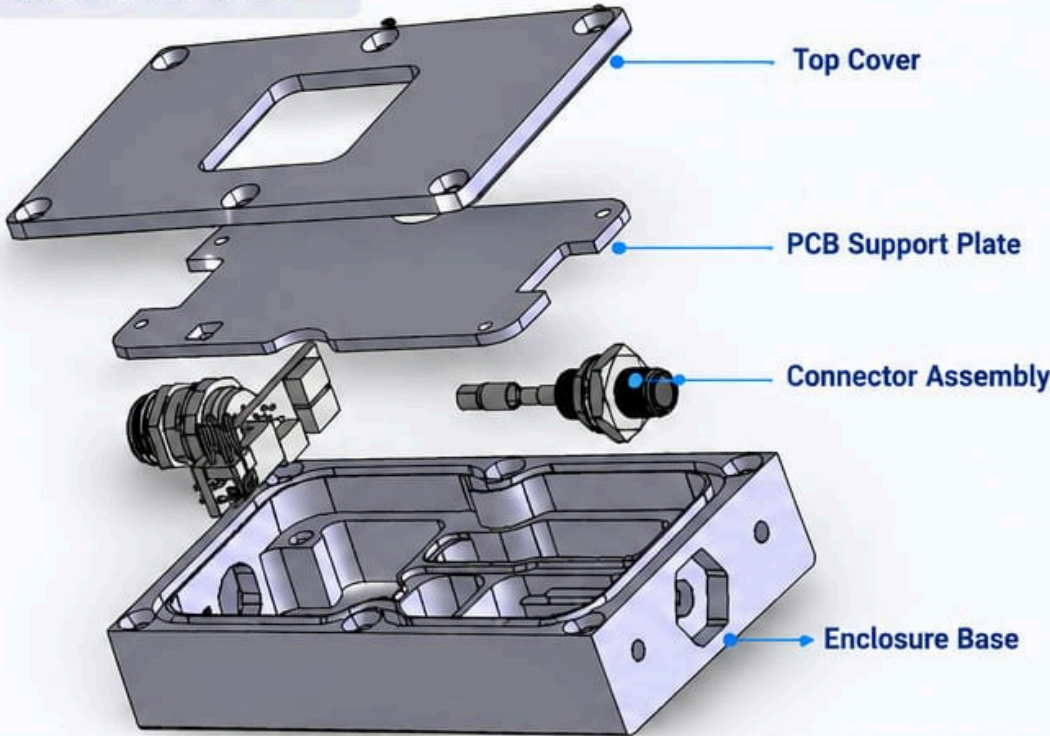
DESIGN OVERVIEW & KEY FEATURES

1. FINAL ASSEMBLY

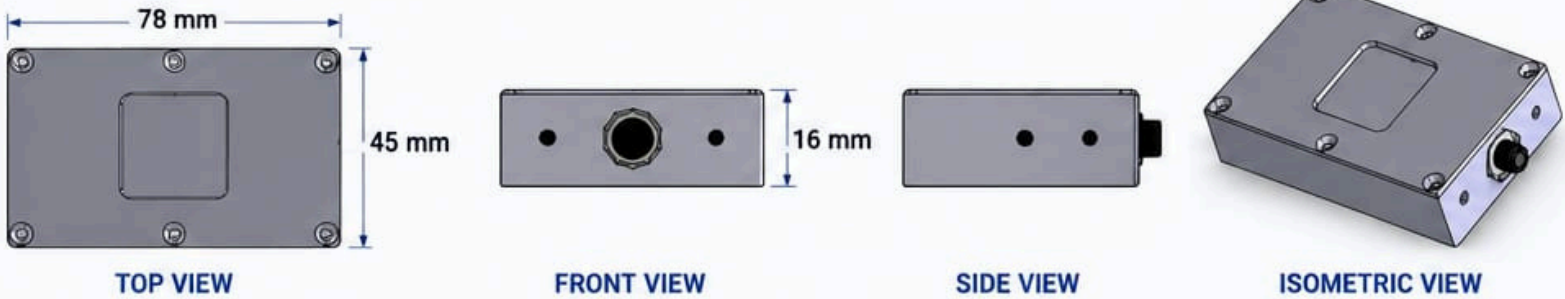


- Material** : Al 6061-T6
- Overall Size** : 78 × 45 × 16 mm
- Key Features :**
- 6 Mounting Screws
 - PCB Cavity
 - Connector Port
 - Compact Design

2. EXPLODED VIEW



3. DIMENSIONAL VIEWS



DESIGN HIGHLIGHTS

- Designed based on engineering drawings
- Accurate enclosure geometry with mounting features
- Integrated PCB, connector and internal support plate
- Optimised for manufacturability and assembly

4. KEY DESIGN FEATURES

- Compact and robust design
- PCB mounting and alignment features
- Connector interface
- Easy assembly and maintenance
- Suitable for thermal analysis and further optimisation

TOOLS USED



CATIA V5
3D Modelling



SpaceClaim
Geometry Editing



Thermal Analysis

01 - Compact Enclosure

Overall envelope: 78 x 45 x 16 mm

02 - Component Integration

Designed to accommodate the PCB, connector interfaces and internal support features.

03 - Mechanical Features

Mounting holes, cover interface, connector opening, power PCB opening and internal supports were incorporated for assembly.

04 - Analysis Preperation

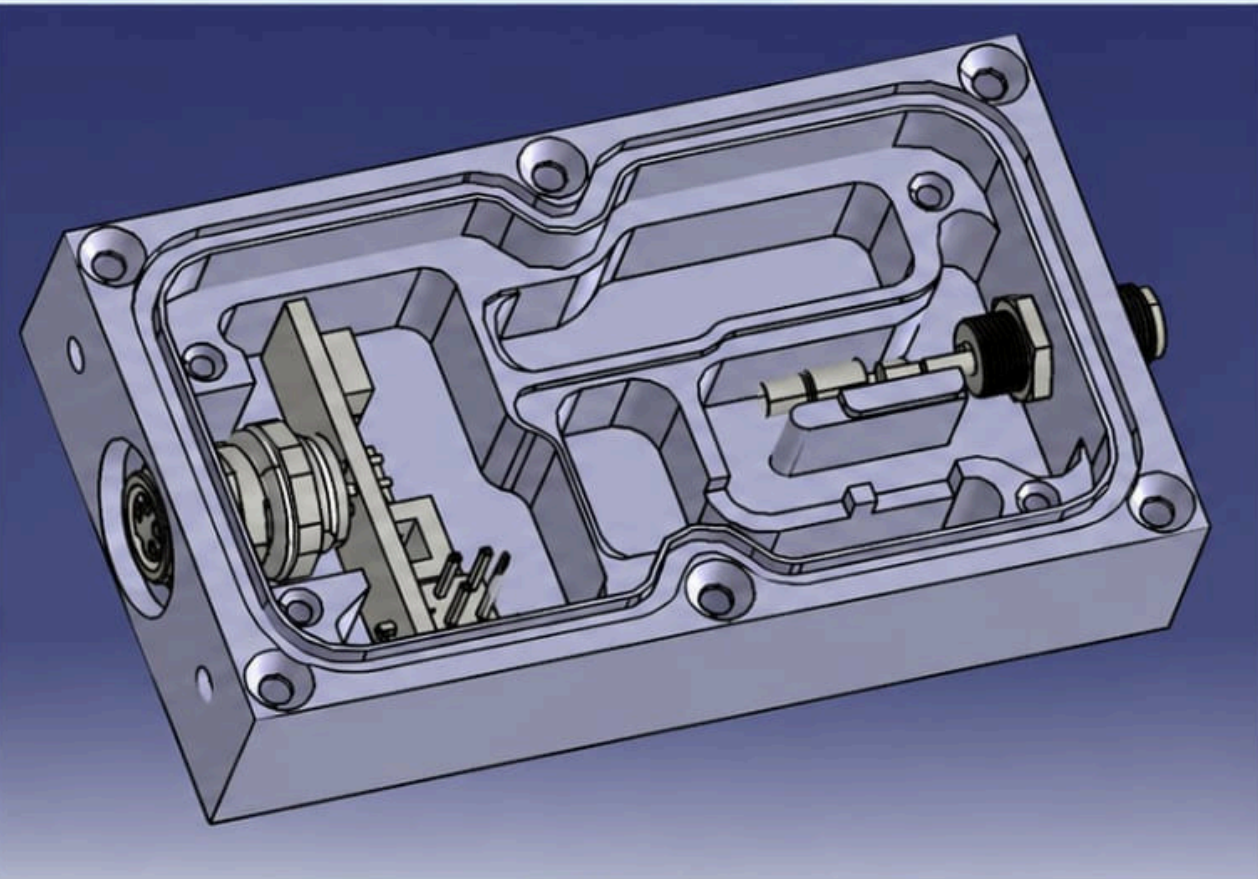
Geometry was developed with consideration for subsequent thermal analysis in ANSYS.

DESIGN WORKFLOW

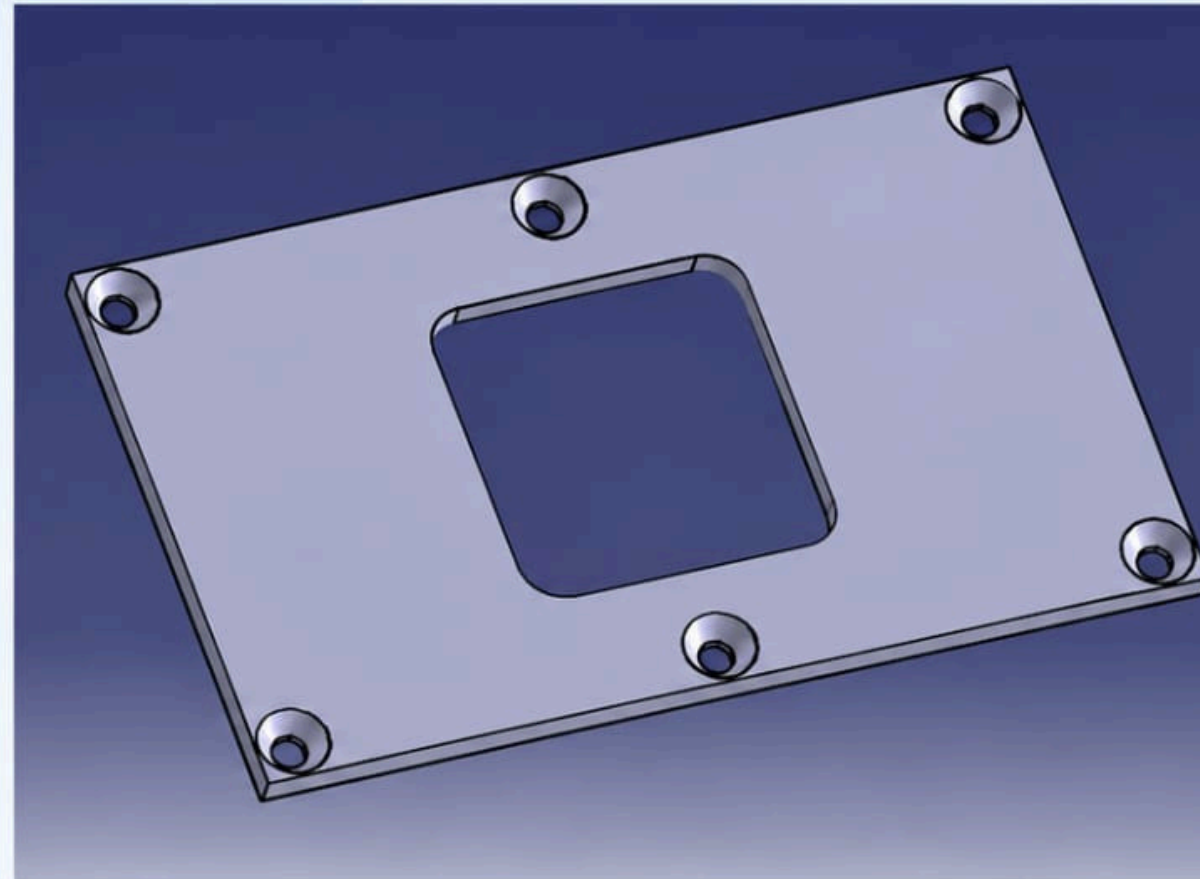
Engineering Drawing → CAD Modelling → Assembly →
Component Integration → Thermal Analysis

CAD MODELLING AND DESIGN DEVELOPMENT

01. ENCLOSURE BASE (INTERNAL VIEW)



02. TOP COVER



KEY CAD FEATURES

1. Overall envelope: 78 x 45x 16 mm
2. Mounting holes and fastener interfaces.
3. Internal PCB support features.
4. Top cover with cut-out.
5. Optimised geometry for assembly and analysis.

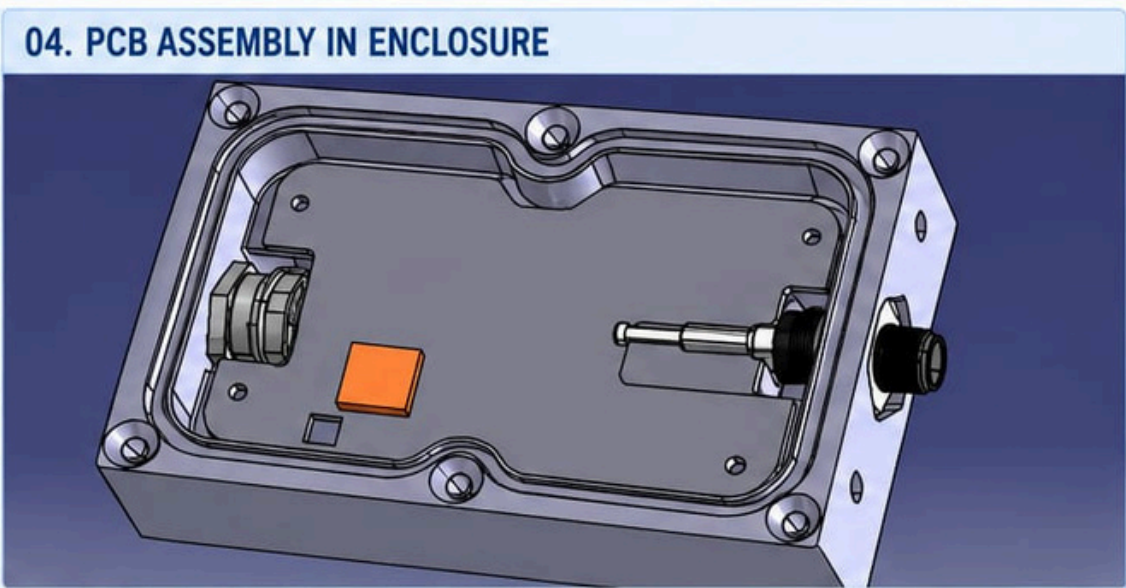
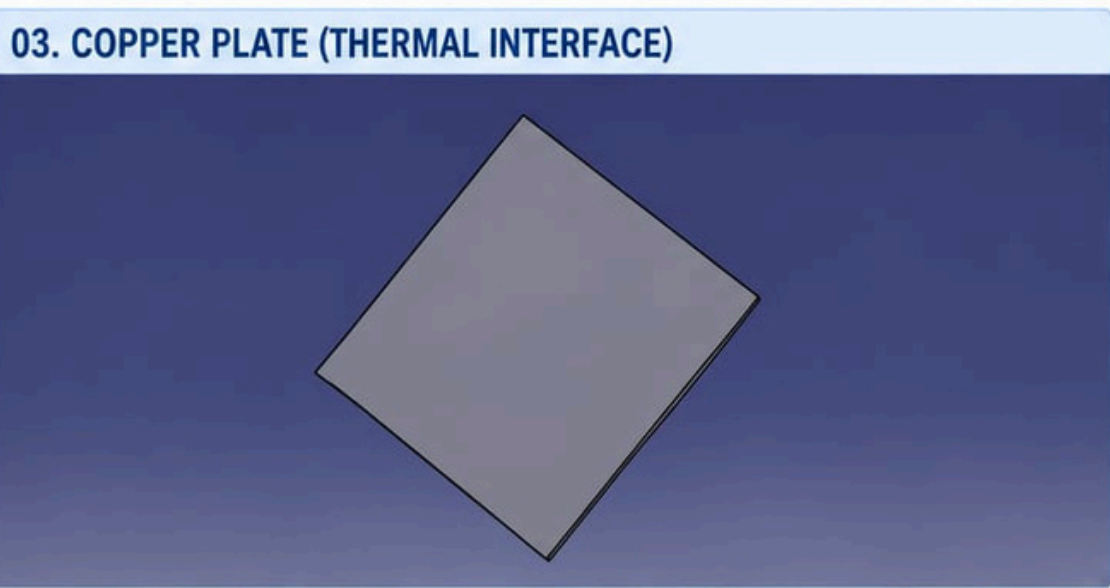
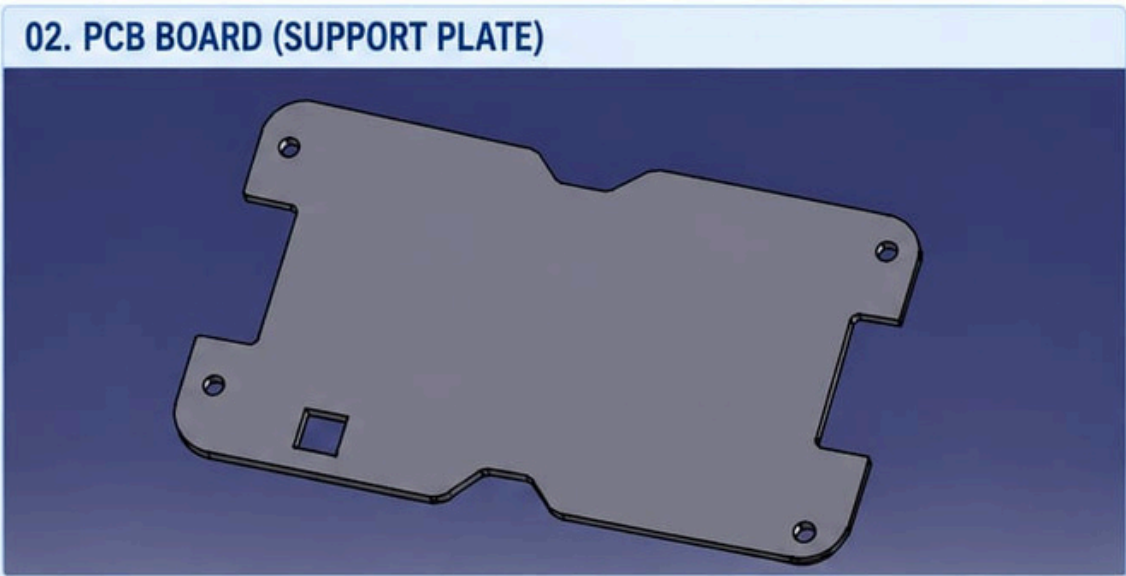
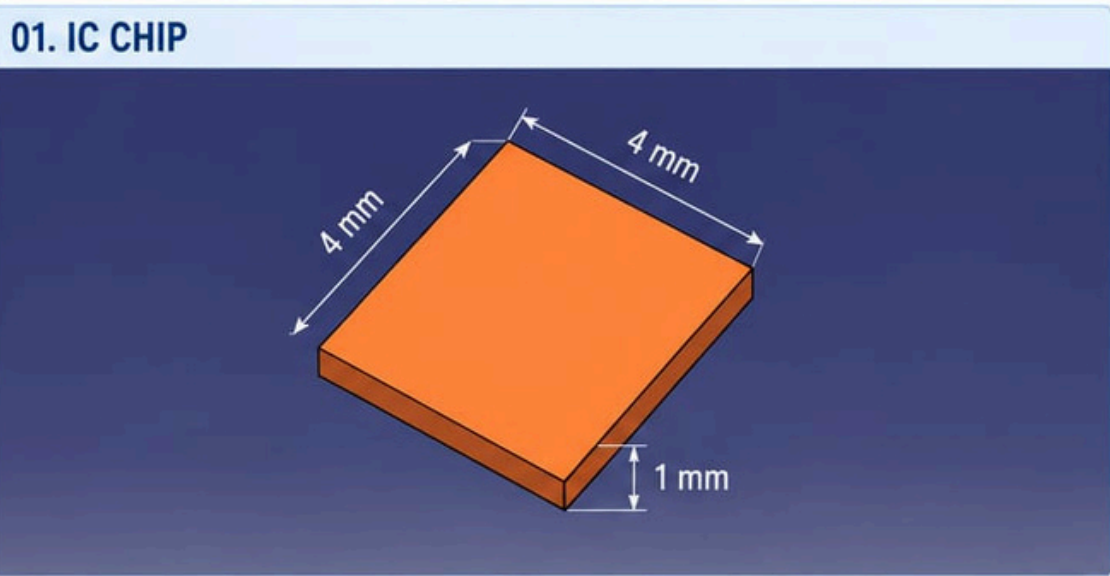
TOOLS USED



Modelling Approach

The enclosure was modelled from the provided engineering drawings using a feature based CAD workflow. The outer envelope was created first, followed by internal pockets, mounting features, connector provision and cover interface. The design was developed with consideration for component integration and subsequent thermal analysis.

PCB & COMPONENT INTEGRATION



TOOLS USED

 CATIA V5
3D Modelling

 SpaceClaim
Geometry Editing

PCB ASSEMBLY IN ENCLOSURE

The PCB support plate is integrated within the enclosure together with the connectors and IC assembly, ensuring proper positioning and assembly access.

IC CHIP

Dimensions: 4 × 4 × 1 mm

The IC chip is positioned on the copper thermal interface to provide a direct heat-transfer path toward the enclosure.

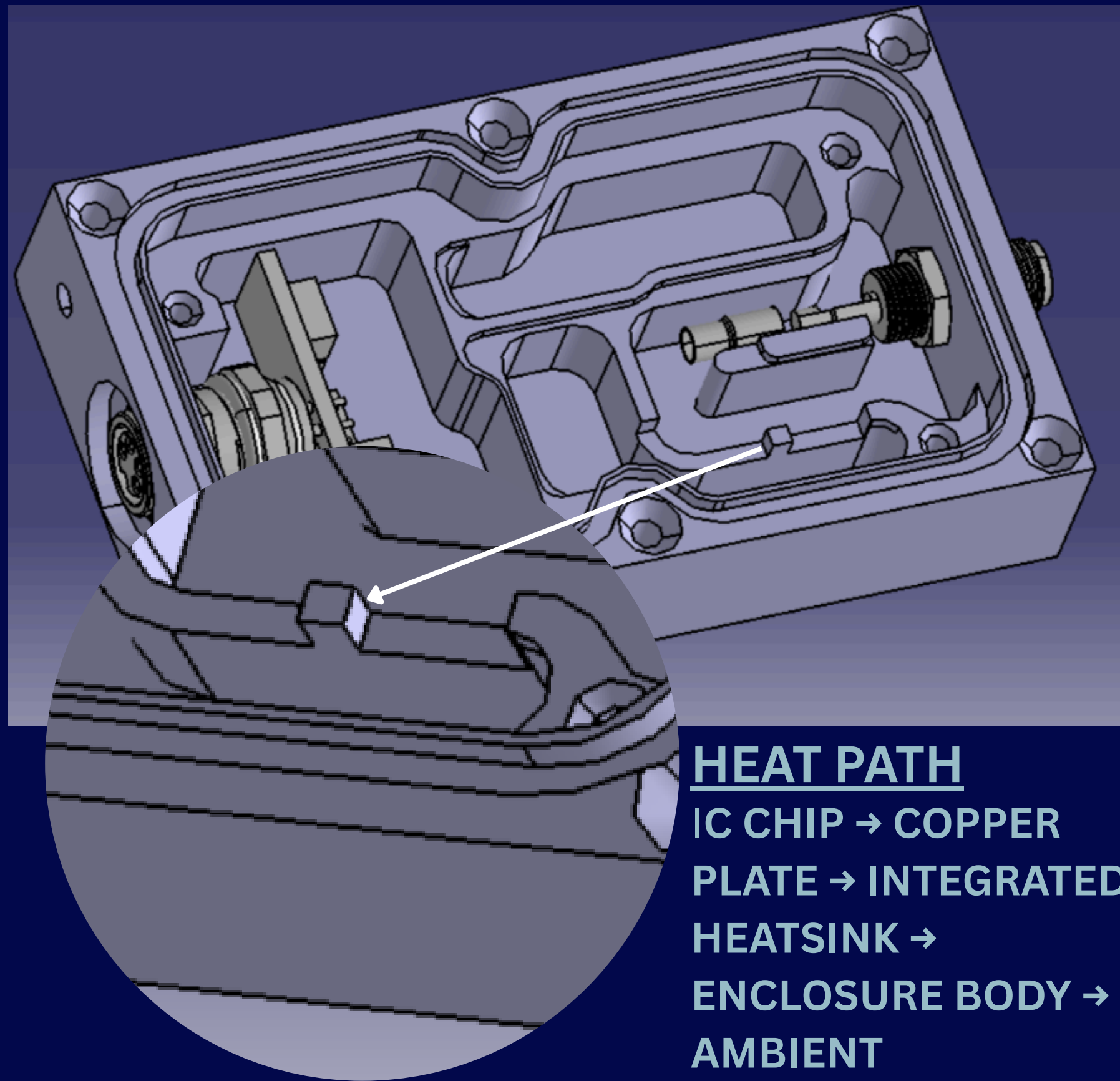
PCB SUPPORT PLATE

Designed to provide a mounting and alignment interface for the PCB while maintaining the required internal clearances.

COPPER PLATE – THERMAL INTERFACE

A copper plate is positioned directly beneath the IC chip to improve heat conduction from the chip toward the heatsink region.

THERMAL MANAGEMENT & INTEGRATED HEATSINK



HEAT PATH

IC CHIP → COPPER
PLATE → INTEGRATED
HEATSINK →
ENCLOSURE BODY →
AMBIENT

HEAT GENERATION

The IC chip is the primary heat-generating component during operation.

COPPER THERMAL INTERFACE

A copper plate is placed directly beneath the IC chip to provide a conductive path for transferring heat away from the chip.

INTEGRATED HEATSINK

The raised extension of the enclosure acts as an integrated heatsink. It is positioned to receive heat from the IC through the copper thermal interface and conduct it into the enclosure body.

HEAT TRANSFER

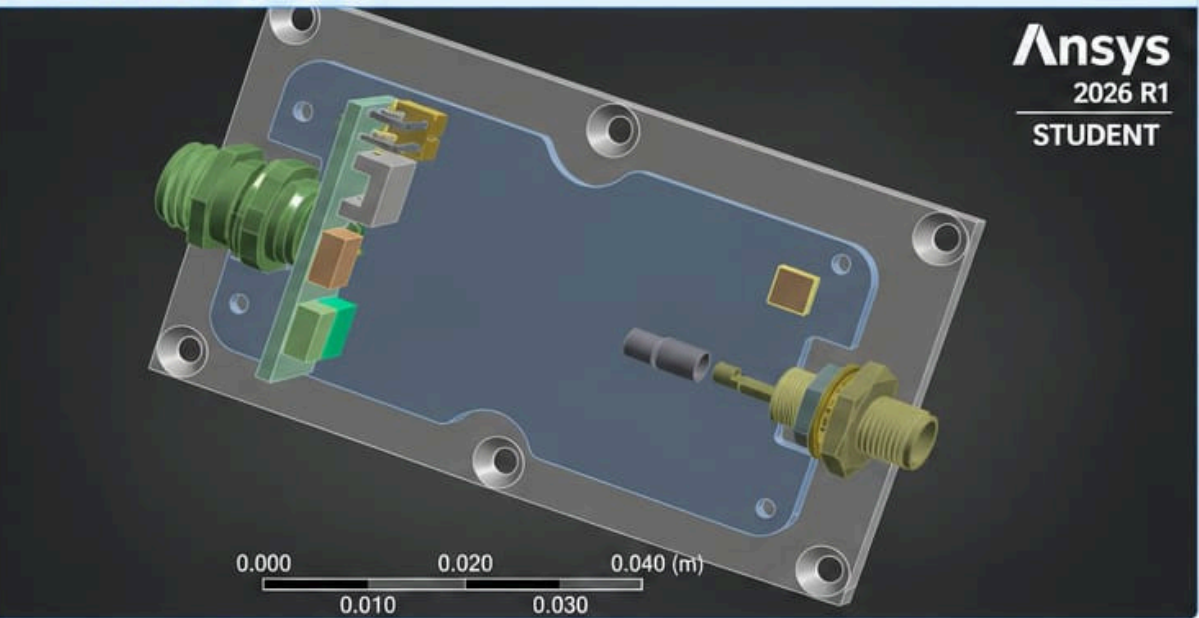
Heat is conducted from the IC chip → copper plate → enclosure extension (heatsink) and then spreads through the surrounding enclosure material.

THERMAL DESIGN

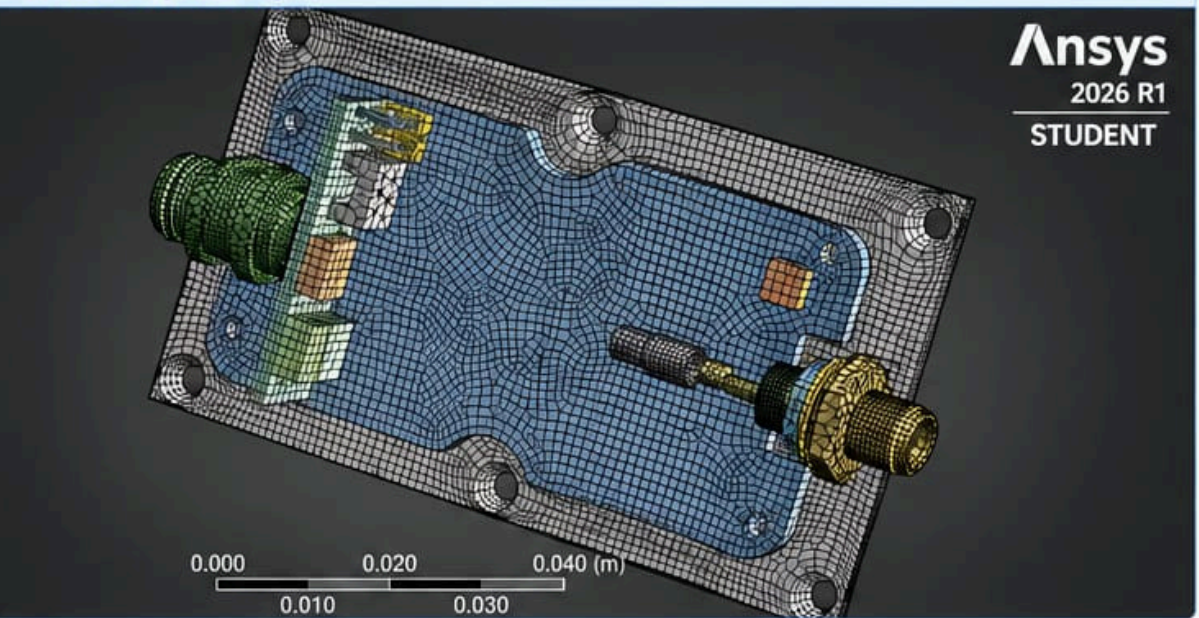
The integrated heatsink eliminates the need for a separate external heatsink and uses the enclosure structure itself as part of the thermal-management system.

THERMAL ANALYSIS

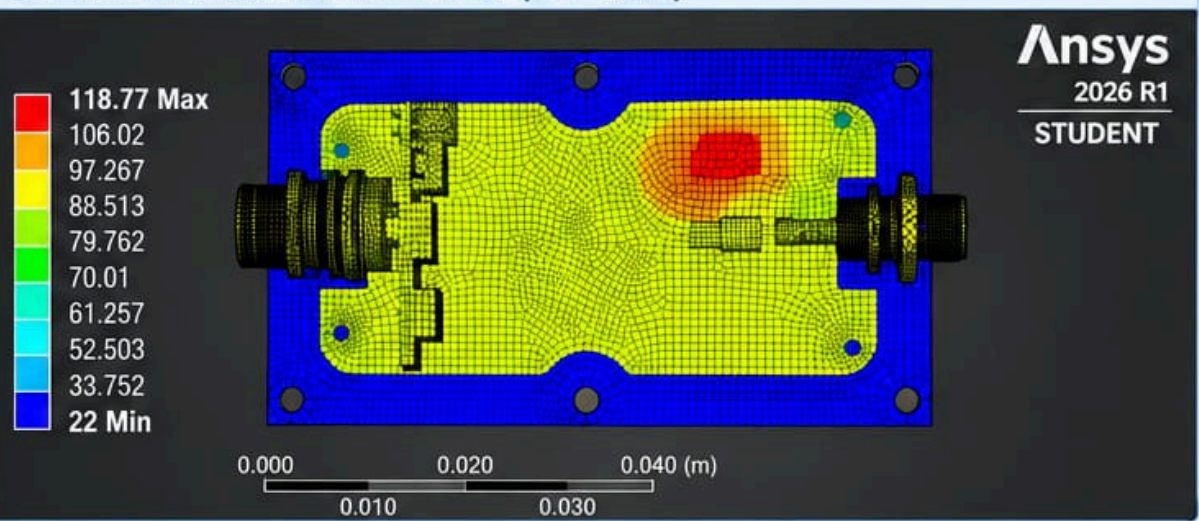
01. GEOMETRY (ANSYS)



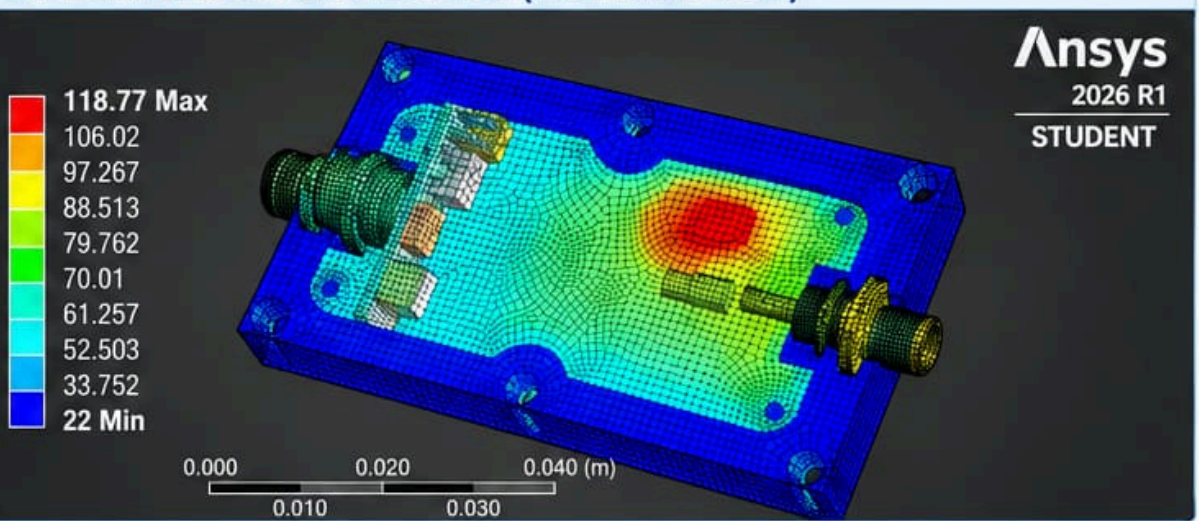
02. MESH (ANSYS)



03. TEMPERATURE DISTRIBUTION (TOP VIEW)



04. TEMPERATURE DISTRIBUTION (ISOMETRIC VIEW)



THERMAL ANALYSIS SETUP

Analysis: Steady-State Thermal

Mesh Size: 1×10^{-3} m (1 mm)

Thermal Contacts: Defined between the relevant component interfaces to establish the heat-transfer path.

Heat Source: Heat generation applied at the IC chip.

Convection: Applied at the external enclosure surfaces for heat dissipation to ambient.

RESULTS

Maximum Temperature: 118.77 °C

Minimum Temperature: 22 °C

Average Temperature: 68.59 °C

The temperature contour shows the highest temperature concentrated around the IC, with heat spreading through the surrounding enclosure. The enclosure surfaces show progressively lower temperatures away from the heat source, indicating heat transfer toward the ambient environment.

PROJECT OUTCOME

DESIGN COMPLETION

Successfully developed the PCB electronics enclosure* from engineering drawings into a complete 3D CAD assembly.

COMPONENT INTEGRATION

Integrated the PCB support plate, IC chip, copper thermal interface, connectors and enclosure features with appropriate mounting and assembly considerations.

THERMAL ANALYSIS

Performed a steady-state thermal analysis in ANSYS to evaluate temperature distribution and heat transfer within the enclosure.

KEY RESULT

The analysis identified a maximum temperature of 118.77 °C at the IC region, with heat spreading through the surrounding enclosure.

ENGINEERING TAKEAWAY

The project provided practical experience in CAD modelling, assembly integration, thermal simulation and engineering analysis.

CATIA V5 | SpaceClaim | ANSYS

CONTACT

Seeking opportunities in additive
Manufacturing, DFAM, Mechanical
Design and Advanced Manufacturing.

Email: [sudeepchowdary27@gmai.com]

Phone: [+91 7816057828]

LinkedIn: [[LinkedIn](#)]

Location: [Hyderabad, India]